PROJECT STATUS REPORT

ClaudeCode Advancements

Research, tools, and systems for AI-assisted development

1,909	40,748	9	241
TESTS PASSING	LINES OF CODE	MODULES	COMMITS

18	306	9	28d
master tasks	intelligence findings	live hooks	project age

Session 57 · 2026-03-19

github.com/mpshields96/ClaudeCodeAdvancements

CONTENTS

Contents

1 Executive Summary
Project overview, health dashboard, and key metrics

2 Five Frontiers
Status of the five core research areas

3 Module Deep-Dives
Detailed breakdown of each module's capabilities and progress

4 Master Tasks
All aspirational goals — complete, active, pending, and blocked

5 Priority Queue
Decay-based priority scoring for active work

6 Live Infrastructure
Hook architecture integrated into Claude Code

7 Intelligence & Research
Reddit findings, academic papers, and self-learning metrics

8 Architecture Decisions
Key technical decisions and their rationale

9 Risks & Blockers
Known issues, mitigations, and technical debt

10 Next Priorities
Upcoming work items ranked by impact

LATEST SESSION HIGHLIGHTS

- ✓ 4 GitHub repo full source code evaluations — Polymarket bot (drawdown heat, 7 Kelly multipliers, ...
- ✓ 18 verified academic papers — Tsang intraday seasonality VALIDATES overnight hypothesis, Baker-Mc...
- ✓ 5 r/algotrading deep-reads — time-window filtering, regime filters (HMM), walk-forward validation...
- ✓ MT-9 autonomous scans — r/Kalshi (50, mostly scams), r/algotrading (30, 6 NEEDLEs), r/polymarket ...
- ✓ ClaudeWatch discovery — MT-1 partial resolution.
- ✓ Priority queue refreshed — MT-10 (18.0), MT-11 (14.0), MT-14/MT-9 (12.0).
- ✓ Cross-chat bridge massive delivery — KALSHI_INTEL.md, CCA_TO_POLYBOT.md, CROSS_CHAT_INBOX.md all ...
- ✓ micro_reflect() built — Mid-session self-learning reflection + 10 tests (self-learning/reflect.py).
- ✓ r/ClaudeCode NEEDLE deep-reads — ClaudeWatch found, spec-system validated by community.

EXECUTIVE SUMMARY

Executive Summary

ClaudeCodeAdvancements is a research and development project building validated tools, hooks, and systems for Claude Code users and AI-assisted development workflows. Launched 2026-02-19 with 57 development sessions completed. All five original research frontiers are production-complete with 1,907 automated tests across 9 modules. The project has expanded into 18 master-level tasks covering autonomous intelligence gathering, self-learning loops, academic research integration, and professional design output. Zero external Python dependencies.

Project Health

TESTS

1,909 / 1,909

47 suites — 100% pass rate

CODEBASE

40,748

Source: 20,939 Test: 19,809

Test-to-source ratio: 0.95:1

VELOCITY

241 commits

8.6/day · 4.2/session

28 days · 57 sessions

MODULES

9

52 source files · 47 test files

0 stubs · 0 syntax errors

MASTER TASKS

18

5 done · 8 active · 4 pending

INTELLIGENCE

306 findings

21 papers · 9 live hooks

Zero external dependencies

FIVE FRONTIERS

Five Frontiers

The five core research areas that drive CCA's development, each validated through community intelligence, Anthropic research, and developer surveys.

1 Persistent Cross-Session Memory

Active CRITICAL

Every session starts from zero — this gives Claude a brain that persists.

TESTS: **132** LOC: **1,259**

2 Spec-Driven Development

Complete HIGH

Unstructured prompting produces poor architecture. Enforce requirements > design > tasks > implement.

TESTS: **90** LOC: **485**

3 Context Health Monitor

Active HIGH

Context rot causes silent output degradation. Monitor, warn, and auto-handoff.

TESTS: **266** LOC: **2,197**

4 Multi-Agent Conflict Guard

Complete HIGH

Parallel agents overwrite each other. Guard credentials, paths, and system modifications.

TESTS: **264** LOC: **1,936**

5 Usage Transparency Dashboard

Complete MEDIUM

No real-time token/cost visibility. Counter, alerts, and structural completeness.

TESTS: **197** LOC: **2,262**

MODULE DEEP-DIVES

Module Deep-Dives

Memory System

Active

TESTS **132** LOC **1,259** FILES **3** `memory-system/`

Persistent cross-session memory. Captures decisions, preferences, and context across Claude Code sessions using hooks and MCP.

KEY COMPONENTS

- Capture hook (PostToolUse + Stop)
- CLI viewer (stats, search, list)
- 8-char hex ID schema
- MCP retrieval server
- Compaction-resistant handoff

Spec System

Complete

TESTS **90** LOC **485** FILES **2** `spec-system/`

Spec-driven development workflow. Enforces requirements > design > tasks > implement sequence with approval gates.

KEY COMPONENTS

- /spec:requirements interview scaffold
- /spec:tasks atomic decomposer
- PreToolUse spec validator hook
- /spec:design architecture generator
- /spec:design-review (4 expert personas)
- UserPromptSubmit auto-activation

Context Monitor

Active

TESTS **266** LOC **2,197** FILES **8** `context-monitor/`

Context health monitoring with adaptive thresholds. Tracks token usage, warns at quality ceilings, blocks exit at critical levels.

KEY COMPONENTS

- PostToolUse token meter
- Stop hook auto-handoff
- Automatic session wrap trigger
- PreToolUse threshold alert
- Compact anchor (every 10 turns)
- Session pacer (2-3h autonomous runs)

Agent Guard

Complete

TESTS **264** LOC **1,936** FILES **7** `agent-guard/`

Multi-agent safety and conflict prevention. Guards credentials, network ports, dangerous paths, and system modifications.

KEY COMPONENTS

- Mobile approver (ntfy.sh push)
- Network/port exposure guard
- Path validator (LIVE in hooks)
- File ownership manifest
- Credential extraction guard
- Content scanner (9 threat categories)
- Session guard (slop detection)

Usage Dashboard

Complete

TESTS **197** LOC **2,262** FILES **5** `usage-dashboard/`

Token and cost transparency. CLI counter, OpenTelemetry receiver, cost alerts, and structural completeness checker.

KEY COMPONENTS

- Token counter CLI
- PreToolUse cost alert hook
- OTLP HTTP/JSON receiver
- /arewedone completeness checker

Reddit Intelligence

Active

TESTS **316** LOC **3,533** FILES **7** `reddit-intelligence/`

Community signal research. Discovers tools, patterns, and pain points from Reddit and GitHub to drive development priorities.

KEY COMPONENTS

- Reddit reader (posts + all comments)
- GitHub repo evaluator
- Subreddit profile registry (10 profiles)
- Autonomous scan pipeline
- Sandboxed repo tester

Self-Learning

Active

TESTS **432** LOC **4,991** FILES **10** `self-learning/`

Cross-session improvement through structured observation, pattern detection, and autonomous proposal generation.

KEY COMPONENTS

- JSONL event journal
- Trace analyzer (4 detectors)
- Sentinel adaptive mutation
- Findings resurfacer
- Pattern reflector + strategy engine
- YoYo improvement loop
- Academic paper scanner
- Skillbook injection hook

NEXT: **Next:** (1) Kalshi chats pick up S55+S56 massive delivery on hotspot. (2) Generate updated CCA report PDF (MT-17). (3) Deep-read remaining 7 r/ClaudeCode NEEDLEs. (4) MT-10 Phase 3: graduate self-learning to Kalshi bot. (5) MT-11: automated GitHub trending scan. (6) Retry blocked URLs on hotspot (SSRN, quantvps).

Design Skills

Complete

TESTS **124** LOC **2,461** FILES **6** `design-skills/`

Professional visual output. Typst-based PDF reports, presentation slides, HTML dashboards, and SVG chart generation.

KEY COMPONENTS

- Report generator (Typst pipeline)
- Dashboard generator (HTML)
- Design guide + visual language
- Slide generator (16:9 PDF)
- Chart generator (SVG)

Research

Complete

TESTS **86** LOC **1,397** FILES **3** `research/`

R&D tools and experimental capabilities including iOS project generation and Xcode build wrapper.

KEY COMPONENTS

- iOS project generator (SwiftUI)
- Xcode build wrapper

MASTER TASKS

Master Tasks

18 tasks defined · **5 complete** · **8 active** · 4 pending · 1 blocked

Completed

5 tasks delivered

MT-2 Mermaid Architecture Diagrams in Spec System

Complete

Phase 1/1

COMPLETE (Session 19). Mermaid diagram requirement added to spec:design at lines 34 and 109.

MT-3 Virtual Design Team Plugin

Complete

Phase 1/1

COMPLETE (Session 21). `/spec:design-review` command created with 4 expert personas (UX, Security, Performance, Maintainability). Consolidated review with APPROVE/REVISE/REDESIGN verdict.

MT-4 Frontend Design Excellence via Design Vocabulary

Complete

Phase 1/1

COMPLETE (Session 21). Section 1b "Design References" added to spec:design — optional for UI features, includes reference UIs, design vocabulary, layout pattern, color constraints. Asks user once if no preferences specified.

MT-6 On-Demand Subreddit Scanner

Complete

Phase 1/1

COMPLETE (Session 27). Delivered:

DELIVERED

- profiles.py: 10 builtin subreddit profiles (4 domains: claude/trading/dev/res...
- --profile and --quick flags added to nuclear_fetcher.py

MT-7 Programmatic Trace Analysis for Self-Learning

Complete

Phase 1/1

COMPLETE (Session 26). Delivered:

DELIVERED

- self-learning/trace_analyzer.py: TranscriptEntry, TranscriptSession, RetryDet...

Active

8 tasks in progress

MT-0 Kalshi Bot Self-Learning Integration

In Progress

Phase 1/2

Phase 1 COMPLETE (Session 21). Trading domain schema built in CCA self-learning module:

DELIVERED

- get_trading_metrics() with PnL tracking, win rate, by-market-type and by-stra...
- Phase 2: Deploy to polymarket-bot (requires cross-project work)

REMAINING

- Phase 2: Deploy to polymarket-bot (requires cross-project work)

MT-9 Autonomous Cross-Subreddit Intelligence Gathering

In Progress

Phase 1/3

Phase 2 COMPLETE (Session 31). Delivered:

DELIVERED

- autonomous_scanner.py: ScanPrioritizer (staleness + yield + diversity scoring...
- execute_scan(): End-to-end pipeline — fetch → filter → dedup → classify → rec...
- ScanResult dataclass with to_dict/save_json for structured output.
- /cca-nuclear autonomous mode: auto-picks highest-priority sub via scanner.
- Phase 3: Supervised trial — run with Matthew monitoring for 3 sessions.

REMAINING

- Phase 3: Supervised trial — run with Matthew monitoring for 3 sessions.

MT-10 YoYo-Pattern Continuous Self-Learning and Self-Building

In Progress

Phase 2/3

Phase 1 COMPLETE (Session 28), QualityGate added (Session 29), E2E validated (Session 29).

DELIVERED

- improver.py: ImprovementProposal lifecycle, ImprovementStore (JSONL), Proposa...
- QualityGate: Geometric mean anti-gaming (Nash 1950 / sentrux pattern). Preven...
- Phase 2 COMPLETE (Sessions 42-44): 5/5 validation sessions, 36 transcripts an...
- Phase 3: Two tracks: (A) Graduate self-learning to Kalshi bot (requires cross...

REMAINING

- Phase 3: Two tracks: (A) Graduate self-learning to Kalshi bot (requires cross-project ...

MT-11 Autonomous GitHub Repository Intelligence

In Progress

Phase 1/2

Phase 1 COMPLETE (Session 30). Delivered:

DELIVERED

- github_scanner.py: RepoMetadata (from GitHub API), RepoEvaluator (0-100 scori...
- Phase 2: Live GitHub API integration (web fetch trending + search).

REMAINING

- Phase 2: Live GitHub API integration (web fetch trending + search).

MT-12 Academic Research Paper Integration

In Progress

Phase 1/2

Phase 1 COMPLETE (Session 38). Delivered:

DELIVERED

- self-learning/paper_scanner.py: Semantic Scholar + arXiv API integration, 4 C...
- Phase 2: Run across all 4 domains, log top papers. Increase delay to 3s (429 ...

REMAINING

- Phase 2: Run across all 4 domains, log top papers. Increase delay to 3s (429 rate limi...

MT–13 iOS App Development Capability

In Progress

Phase 1/3

Phase 2 COMPLETE (Session 49). Delivered:

DELIVERED

- ios_project_gen.py: Generates complete Xcode projects from CLI (SwiftUI + tes...
- xcode_build.py: Python xcodebuild wrapper (build, clean, test, scheme/simulat...
- Phase 3: Build first real app (Kalshi mobile dashboard) or install SwiftUI Ag...

REMAINING

- Phase 3: Build first real app (Kalshi mobile dashboard) or install SwiftUI Agent Skill

MT–14 Autonomous Re-Scanning of Previously Scanned Subreddits

In Progress

Phase 1/3

Phase 1 COMPLETE (Session 35). Delivered:

DELIVERED

- rescan_sub() method on AutonomousScanner: delta-scanning with timestamp filte...
- get_stale_subs() helper for identifying overdue subs
- Phase 2: Run rescan on a real stale sub, validate delta filtering works. Phas...

REMAINING

- Phase 2: Run rescan on a real stale sub, validate delta filtering works. Phase 3: Wire...

MT–17 UI/Design Excellence and Professional Report Generation

In Progress

Phase 4/5

Phase 1 COMPLETE (Session 41). Delivered:

DELIVERED

- design-skills/ module with CLAUDE.md, design-guide.md (color palette, typogra...
- design-skills/report_generator.py — CCADataCollector + ReportRenderer + CLI (...
- design-skills/templates/cca-report.typ — Professional status report template
- /cca-report slash command for one-command PDF generation
- Phase 2 COMPLETE (Session 47): slide_generator.py + /cca-slides command (25 t...
- Phase 3 COMPLETE (Session 48): dashboard_generator.py + /cca-dashboard comman...
- Phase 4 COMPLETE (Session 49): chart_generator.py — pure SVG chart generation...
- Phase 5: Website templates — landing pages, documentation sites

REMAINING

- Phase 5: Website templates - landing pages, documentation sites
- Phase 5: Website templates — landing pages, documentation sites

Pending & Blocked

5 tasks awaiting start or unblocked

MT-5 Claude Pro ↔ Claude Code Bridge

Not Started

Future task. Needs research on Claude Pro's current integration options.

MT-1 Maestro-Style Visual Grid UI

Blocked

Blocked on macOS 15.6 beta SDK. Check for new releases each session.

MT-8 iPhone Remote Control Perfection

Not Started

Not started. Needs r/ClaudeCode + r/ClaudeAI research sweep for remote control posts.

MT-18 Academic Writing Workspace

Not Started

Not started. ClaudePrism repo: github.com/delibae/claude-prism. Session 39 created this task.

MT-19 Local LLM Fine-Tuning Capability

Not Started

Phase 3/5

Not started. Long-term exploration. Unsloth repo: github.com/unslothai/unsloth.

REMAINING

- Phase 5: website templates. Report v2 built S54. |

PRIORITY QUEUE

Priority Queue

Active tasks ranked by decay-based priority scoring. Score = base value + aging penalty. Higher = work on this first.

1	MT-10	**YoYo self-learning** Phase 3A: Resurfacer done (S46). Phase 3B: graduate to Kalshi bot. Cap: 18.0	18
2	MT-11	**GitHub intelligence** Phase 2 validated. Phase 3: automated trending scan. Cap: 14.0	14
3	MT-14	**Rescan stale subs** All 15 subs scanned. Tighten investing classifier. Cap: 12.0	12
4	MT-9	**Autonomous scanning** Phase 3: all NEEDLEs reviewed. Next: production autonomous runs.	12
5	MT-8	**iPhone remote control** SOLVED by native Remote Control. Cap: 10.0. Consider closing.	10
6	MT-12	**Academic papers** 12 papers delivered S52. Phase 3: domain expansion.	10
7	MT-18	**Academic writing** Research phase. Cap: 8.0	8
8	MT-13	**iOS/macOS app development** Phase 2 COMPLETE. Phase 3: first real app.	7.5
9	MT-17	**Design/reports** Phase 4 COMPLETE. Phase 5: website templates. Report v2 built S54.	7

LIVE INFRASTRUCTURE

Live Infrastructure

9 hooks integrated into Claude Code via settings.local.json. Every tool call passes through this pipeline.

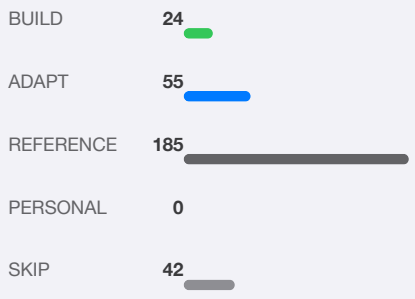
Event	Matcher	Hook	Purpose
`PostToolUse`	*	meter.py	Token counter
`PostToolUse`	*	compact_anchor.py	Context anchor writes
`PreToolUse`	*	alert.py	Warn/block at red/critical
`PreToolUse`	*	cost_alert.py	Cost threshold warning
`PreToolUse`	*	path_validator.py	Dangerous path detection
`PreToolUse`	Bash	credential_guard.py	Credential extraction guard
`UserPromptSubmit`	*	skill_activator.py	Spec auto-activation
`UserPromptSubmit`	*	skillbook_inject.py	Strategy injection
`Stop`	*	auto_handoff.py	Block exit at critical

INTELLIGENCE & RESEARCH

Intelligence & Research

Reddit Intelligence

FINDINGS BY VERDICT



7 subreddits · 30 repos evaluated

Self-Learning System

Strategies	10 total (4 confirmed)
Proposals	6 (all LOW risk)
Trace Sessions	31 analyzed
Avg Score	70 / 100
Papers Logged	21
Sentinel Rate	5-10%

YoYo loop: observe → detect → hypothesize → build → validate

ARCHITECTURE DECISIONS

Architecture Decisions

Decision	Rationale
Local-first storage	User owns all data, no cloud dependency
Stdlib-first (no pip)	Zero dependency management overhead
Hook-based delivery	Native Claude Code integration points
Adaptive context thresholds	Fixed percentages break at 1M windows
JSONL for all logging	Append-only, grep-friendly, no corruption risk
QualityGate geometric mean	Prevents Goodhart's Law gaming of metrics
Whitelist-first phishing detection	Zero false positives on legitimate domains
Typst over WeasyPrint	Single binary, millisecond compile, JSON-native

RISKS & BLOCKERS

Risks & Blockers

MT-1 Maestro Visual Grid UI

BLOCKER

Blocked on macOS 15.6 beta SDK crash. Tauri/React app crashes on launch.

Mitigation: Using tmux + dev-start script as workaround. Waiting for stable macOS release.

Semantic Scholar Rate Limiting

RISK

429 errors at 1.5s delay between queries for academic paper scanning.

Mitigation: Increased to 3s delay with exponential backoff.

Context Burn from File-Writing Hooks

DEBT

compact_anchor.py writes trigger system-reminder context consumption.

Mitigation: Writes limited to every 10 turns, files kept small.

NEXT PRIORITIES

Next Priorities

1. **Kalshi chats pick up S55+S56 massive delivery on hotspot**
2. **Generate updated CCA report PDF (MT-17)**
3. **Deep-read remaining 7 r/ClaudeCode NEEDLEs**
4. **MT-10 Phase 3**
   graduate self-learning to Kalshi bot

5 **MT-11**
automated GitHub trending scan

6 **Retry blocked URLs on hotspot (SSRN, quantvps)**

STATUS

All Systems Operational

1,909 tests · 40,748 LOC · 57 sessions · 241 commits

9 modules · 18 master tasks · Zero external dependencies

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